Mini Climate Emulation Challenge

Nathan Mankovich

Image Processing Lab, University of Valencia, Spain

Why this matters + problem statement

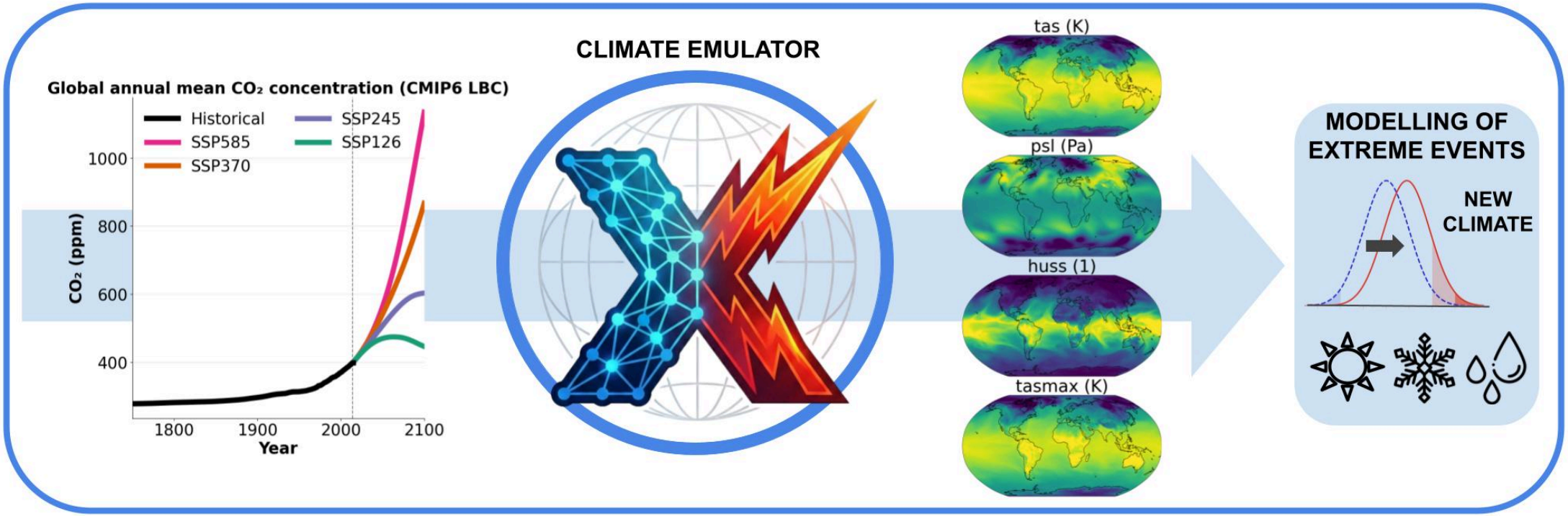
From the challenge brief:

- Extreme events are rising in impact
- Full Earth System Models are expensive to run at scale
- Emulators are needed for fast, scenario-aware analysis

Task: learn a mapping from forcings + historical context to daily climate outputs that preserve behavior relevant to extremes.

$$x_{present} = G(x_{past}, f_{past}, f_{present})$$

What is modeled



Two valid tracks

Track A — Pure ML

- Improve architecture, losses, calibration, compute efficiency
- Output: stronger predictive model and ablation evidence

Track B — Science-driven

- Study where/why extremes are missed (region, variable, index)
- Output: interpretable diagnostics and model-guiding insights

Both paths are valid and can converge into one submission strategy.

Weekly plan

Days 1–2: familiarize with code, brainstorm, & pick a track

Days 3–4: writing code and running experiments

Day 5: prepare & present progress

(Optional) converge into a submission-ready experiment plan for ClimX.

Data introduction

Fig. 1 Overview of SSPs

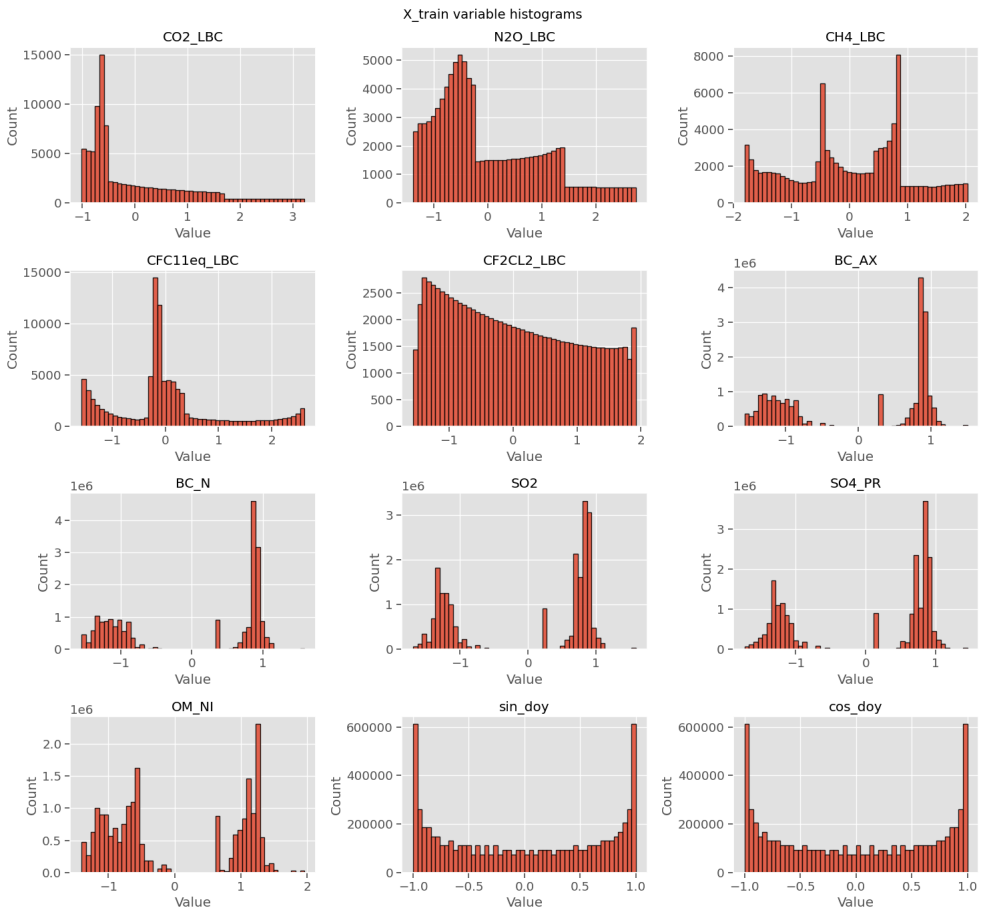
(Narratives in O'Neill et al., 2016, Glob Env Change, online first)

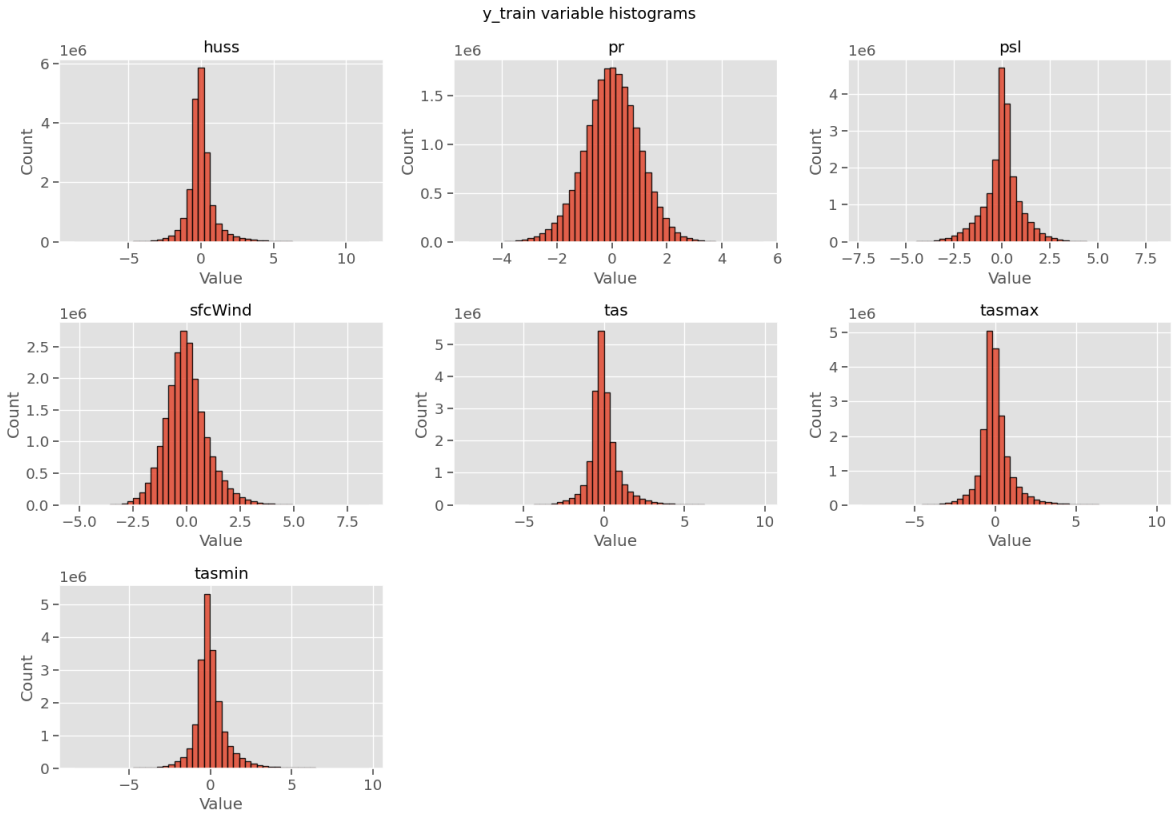
SSP1: low challenges for mitigation (resource efficiency) and adaptation (rapid development)
SSP3: high challenges for mitigation (regionalized energy / land policies) and adaptation (slow development)
SSP4: low challenges for mitigation (global high tech economy), high for adapt. (regional low tech economies)
SSP5: high challenges for mitigation (resource / fossil fuel intensive) and low for adapt. (rapid development)

Inputs: 5 greenhouse gasses (eg., CO2) and 5 aerosols (e.g., SO4)

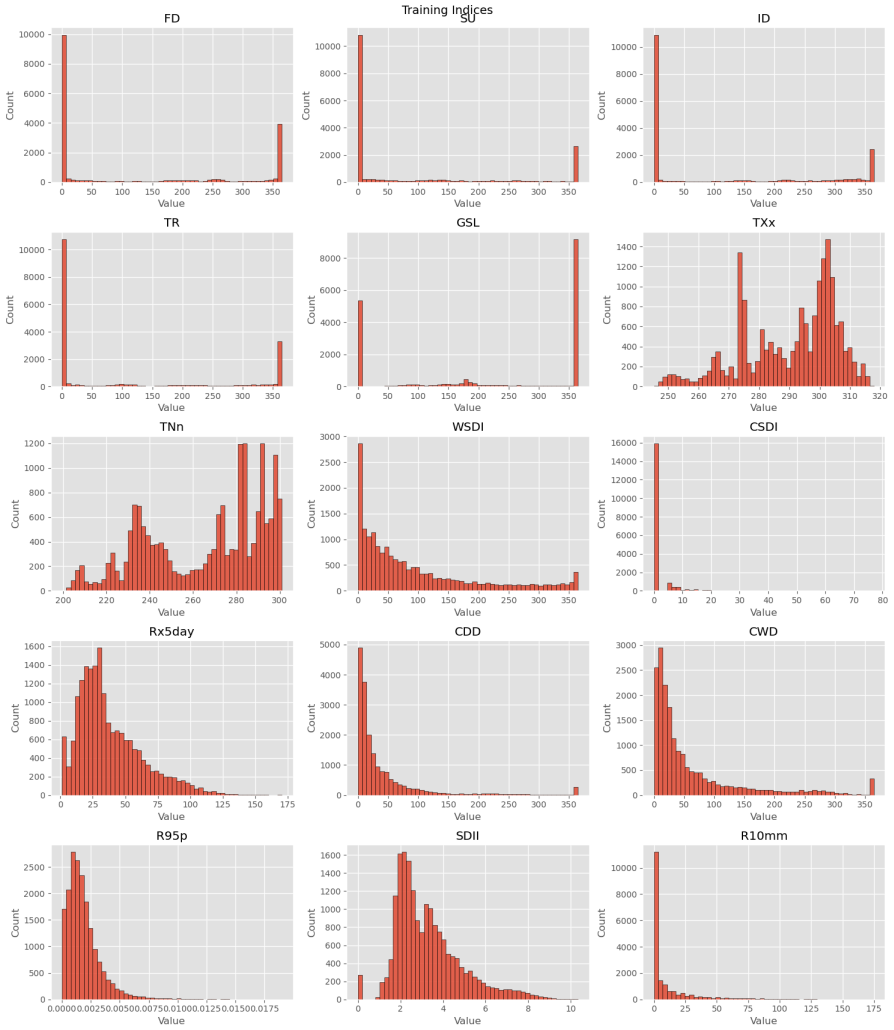
Outputs: 7 climate variables and/ or 15 extreme indices (e.g., ice days)

Inputs & Outputs (SSPs 126, 370, 585)

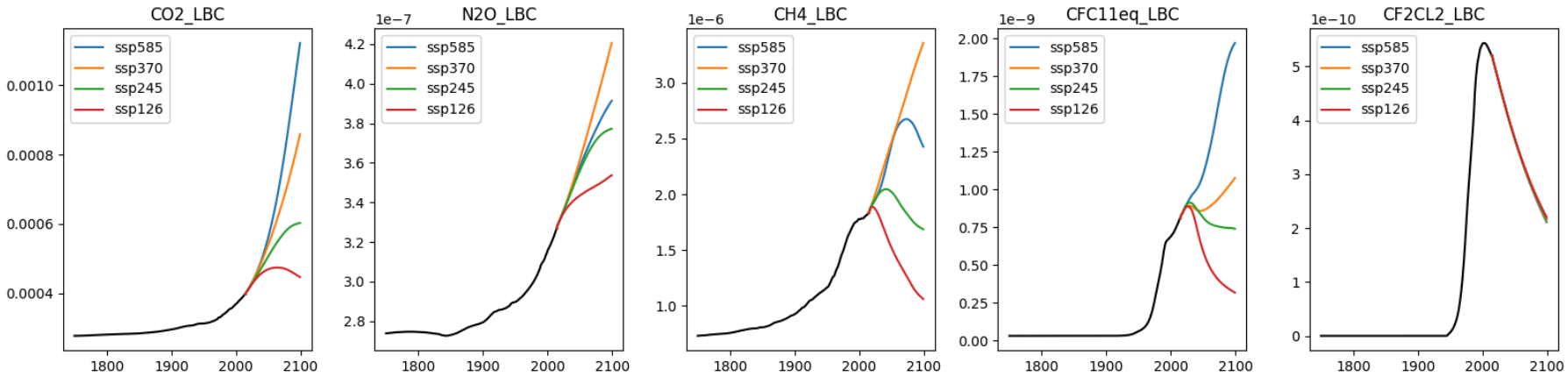


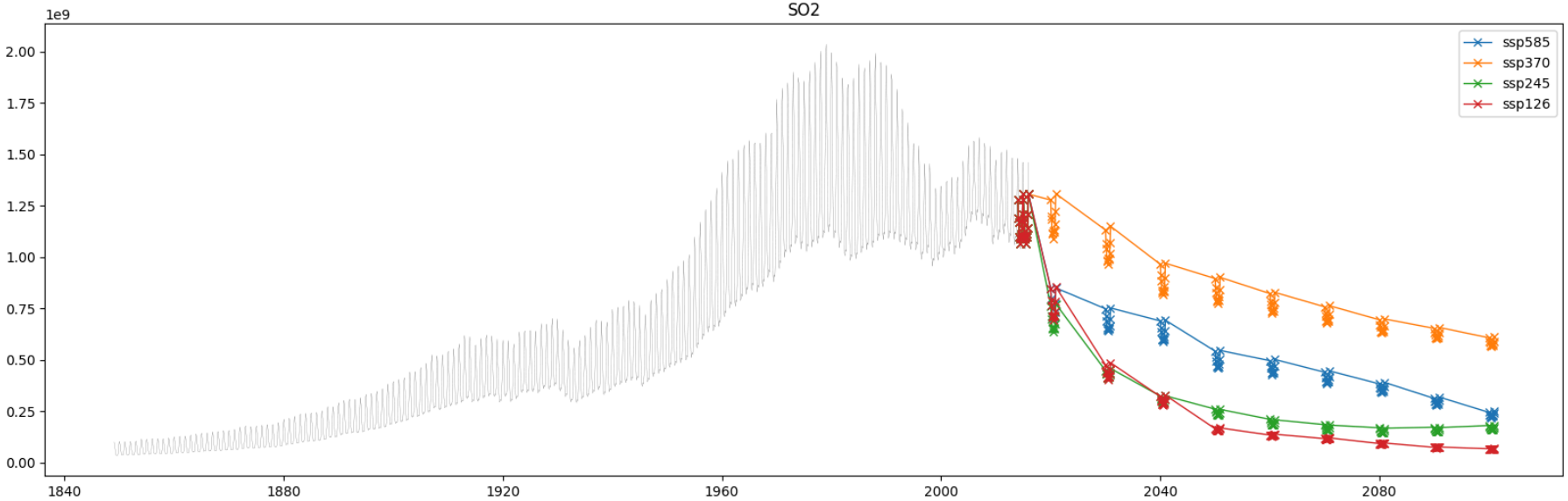


Climate Variables & Indices (SSPs 126, 370, 585)



GHG and Aerosols





Backup: Training & Evaluation

Train/ test split

- Train on historical + selected SSP trajectories
- Validation on late years of training scenarios or hold one scenario out

Evaluation

- Prioritize region-aware extreme-event realism
- Don’t optimize only average-state behavior
- Check robustness across scenarios and seeds
- Track both skill and computational efficiency

Backup: Open Questions

1. How should we handle non-normal or long-tailed distributions in the input variables?
2. What is the optimal train/ validation split? Should historical data be included in training?
3. Which model class gives the best accuracy–compute trade-off for this task?
4. Should we predict the evaluation indices directly, or predict full climate fields and compute indices afterward?
5. How can we scale the same training pipeline from the lite dataset to the full dataset?
6. Which variables and regions contribute most to model error, and can targeted weighting help?
7. How sensitive are results to lead time (near-term vs long-term horizons) across SSP scenarios?
8. Do we need variable-specific loss weighting so low-variance targets are not overshadowed?
9. How stable are results across random seeds and initialization choices?
10. Which calibration method best improves probabilistic reliability of predictions?

Backup: ClimX Details

Rules

- Use only provided challenge data
- Submit through official platform workflow
- Top entries may require reproducibility/code validation
- Open-source release requirements apply for prize eligibility

Timeline

- Launch: mid-April 2026
- Submission deadline: mid-July 2026
- Winners announced: late July 2026